AQ7.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

If $||a||$ = 3||a|| = 3, $||b||$ = 2$\sqrt{2}$||b|| = 2 $\sqrt{2}$ and angle between aa and bb is 45° 45∘, then the value of $a.b$a.b is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point
1 point
Given that $\| a \|$ = 3‖a‖ = 3 and $\| b \|$ = 5‖b‖ = 5 and $a \cdot b$ = 7.5a · b = 7.5, for two vectors aa and bb in R^n Rn. Find the angle (in degrees) between the two vectors a, ba, b. Note: cos60° = 1/2, cos30° = $\sqrt{3}$/2cos 60∘ = 1/2, cos 30∘ = $\sqrt{3}$/2.

- 30° 30∘
- 60° 60∘
- 120° 120∘
- 150° 150∘

No, the answer is incorrect.
Score: 0
Accepted Answers:
60° 60∘
1 point
Consider two vectors a = (3, 4, − 1), b = (2, − 1, 2)a = (3, 4, −1), b = (2, −1, 2) in R^3 R3. Choose the correct options.

- [ ] Length/Norm of aa is $\sqrt{26}$ $\sqrt{26}$ and that of bb is 33.
- [ ] Length/Norm of aa is 55 and that of bb is 33.
- [ ] Length/Norm of aa is $\sqrt{24}$ $\sqrt{24}$ and that of bb is $\sqrt{7}$ $\sqrt{7}$.
- [ ] Angle between the two vectors is 9090 degrees.

☐
Angle between the two vectors is approximately 00 degrees.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Length/Norm of aa is $\sqrt{26}$ 26
$\sqrt{}$
and that of bb is 33.
Angle between the two vectors is 9090 degrees.

1 point

If the dot product of two vectors is zero, then choose the correct option.

○ The two vectors are perpendicular to each other.
○ The two vectors are parallel to each other.
○ The two vectors are exactly the same.
○ The length of the two vectors must be the same.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The two vectors are perpendicular to each other.

Level 2

If a = (6, − 1, 3), b = (4, c, 2)a = (6, −1, 3), b = (4, c, 2) where $a, b \in R^3$ a, b ∈ R3 and aa and bb are perpendicular to each other, then find the value of cc?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 30

1 point

1 point

Consider two vectors a = (1, 2), b = (2, 2)a = (1, 2), b = (2, 2) and θθ is the angle between them. The sum of the two vectors is given by c = a + bc = a + b. Choose the correct options.

☐
Length/Norm of cc is 55.
☐
Length/Norm of cc is 2525.
☐
Length/Norm of cc is 44.
☐
cosθ = $\frac{3}{\sqrt{10}}$cos θ = 10
$\sqrt{}$
3.
☐
cosθ = $\frac{3}{\sqrt{5}}$cos θ = 5
$\sqrt{}$
3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Length/Norm of cc is 55.
$\cos\theta = \frac{3}{\sqrt{10}}$cos θ = 10

√

3.
1 point
Consider 33 vectors a, b, ca, b, c in R^3 R3 and a scalar λλ in RR. Choose the set of correct options.
Note: (.)(.) represents the dot product.

☐ $\lambda(a \cdot b) = (\lambda a) \cdot b$λ(a · b) = (λa) · b

☐ $\lambda(a \cdot b) = (\lambda b) \cdot a$λ(a · b) = (λb) · a

☐ $\lambda(a \cdot b) = (\lambda a) \cdot (\lambda b)$λ(a · b) = (λa) · (λb)

☐ $(a + c) \cdot b = a \cdot b + c \cdot b$(a + c) · b = a · b + c · b

☐ $(a + c) \cdot b = a \cdot c + c \cdot b$(a + c) · b = a · c + c · b

☐ $a \cdot a = 0, b \cdot b = 0$a · a = 0, b · b = 0 if and only if a, ba, b are null vectors.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lambda(a \cdot b) = (\lambda a) \cdot b$λ(a · b) = (λa) · b
$\lambda(a \cdot b) = (\lambda b) \cdot a$λ(a · b) = (λb) · a
$(a + c) \cdot b = a \cdot b + c \cdot b$(a + c) · b = a · b + c · b
$a \cdot a = 0, b \cdot b = 0$a · a = 0, b · b = 0 if and only if a, ba, b are null vectors.

1 point
Consider two vectors $a = (1, 3, 5, 7, 9)$a = (1, 3, 5, 7, 9) and $b = (2, 4, 6, 8, 10)$b = (2, 4, 6, 8, 10). Choose the set of correct options.

☐ The length of bb is more than aa.

☐ The length of aa is more than bb.

☐ The length of $(a - b)$(a − b) is $\sqrt{5}$ 5

√

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☐ The length of $(a - b)$(a − b) is $\sqrt{50}$ 50

√

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No, the answer is incorrect.
Score: 0
Accepted Answers:
The length of bb is more than aa.
The length of $(a - b)$(a − b) is $\sqrt{5}$ 5

√

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Check Answers

Your score is: 0/8